Optimal Reconfiguration of Systems in Aeronautics

Lecture 04: "Fault Tolerant Control"

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2023-24

Outline

1 Fault Tolerant Control

Title I

This chapter introduces the basics of fault tolerant control.

Cuprins

1. Fault Tolerant Control
   - Post-Fault Operation
   - System Reconfiguration and Fault Accommodation
   - Fault Recoverability, System Supervision, and System Reliability
   - Fault Tolerant Control Methods
   - Characterization of Active Reconfiguration Methods

Post-fault operation I

In this respect, remind the general statements of the Standard, Robust, and Adaptive Control Problems presented before and note that, whether during operation there are any faults which affect the system behavior as much as any control law produced as solution of the SCP, RCP, or ACP is not satisfactory, then the control problem under actual operating conditions should be regarded from a different perspective, which is that of fault tolerance.

Thus, the interpretation of fault tolerance w.r.t. the control action of an automatic control system can be given as in the following definition.

Definition (Fault Tolerance)

The fault tolerance is defined as the ability of a controlled system to maintain control objectives despite the occurrence of a fault, whereas a degradation of control performance may be accepted. □

Post-fault operation II

Hence, reconsider the control objective O and the nominal system under unimpaired constraints (S_n, θ_n); moreover, let (S_f, θ_f) be the actual constraints, with $(\hat{S}_f, \hat{\theta}_f)$ the estimated ones.

Obviously, nominal control solves $\langle O_n, S_n, \theta_n, \mathcal{U} \rangle$.

When a fault occurs, $(S_f, \theta_f) \neq (S_n, \theta_n)$, and the nominal control might be no longer suitable; of course, this is a generalization of the robust (RCP) and adaptive (ACP) control problems, since both the parameters and the structure of constraints may change when faults occur.

Therefore, the Fault Tolerant Control Problem (FTCP) can be introduced as in the following definition.

Definition (FTCP - Fault Tolerant Control Problem)

Solve

$$P_{c_f}^{ft} \triangleq \langle O_n, \hat{S}_f, \hat{\Theta}_f, \mathcal{U} \rangle \,, \tag{1}$$

where $(\hat{S}_f, \hat{\Theta}_f)$ is the set of possible structures and parameters of the faulty system. □

Post-fault operation III

In the previous definition, the case where FDIE is not able to estimate the model of the faulty system but only a class of models is presented; for example, the fault cannot be really isolated but a set of possibilities still exist.

Preliminaries I

In any case, it is clear that, if only the fault detection functionality is available, nothing can be done.

However, where diagnosis is available, the set $(\hat{S}_f, \hat{\Theta}_f)$ could be provided by a diagnosis task.

System reconfiguration I

In this respect, if only fault detection and fault isolation functionalities are available, system reconfiguration must be used; it is known which component is faulty but its new model is not known, hence the only thing that can be done is to switch it off and to replace it by another healthy component or use the remaining components by affecting the control computations and the power distribution.

That is, system reconfiguration means to find a pair $(\tilde{S}_f, \tilde{\theta}_f)$ among all feasible pairs $(S_f \times \Theta_f)$ such that $\langle O_n, \tilde{S}_f, \tilde{\theta}_f, \mathscr{U} \rangle$ has a solution; in this situation, a pair $(\tilde{S}, \tilde{\theta})$ is considered feasible if it belongs to the fault-free parts of the faulty system; the fault diagnosis module may also identify a set $(\hat{S}_f, \hat{\Theta}_f)$ to give an estimate of the constraints of the faulty system.

System reconfiguration II

Then, the System Reconfiguration Problem (SSP) can be stated as in the following definition.

Definition (SRP - System Reconfiguration Problem)

Find a new set of system constraints

$$(\tilde{S}_f, \tilde{\theta}_f) \in (S_f, \Theta_f) \text{ or } (\hat{S}_f, \hat{\Theta}_f)\,,$$

such that the control problem

$$P^{sr}_{c_f} \triangleq \langle O_n, \tilde{S}_f, \tilde{\theta}_f, \mathscr{U} \rangle \tag{2}$$

has a solution and compute / activate it. □

Because the hardware architecture of the system is changed, system reconfiguration includes the selection of a new control configuration where alternative input and output signals are used; regarding this, the choice of a new set of constraints will imply that input-output relations between the controller and the plant are changed, as well.

Fault accommodation I

The only case where real fault accommodation can be applied, i.e., using the faulty component, is when its model is available, either because the fault has been identified or because it is learnt through a kind of adaptation scheme.

Thus, the Fault Accommodation Problem (FAP) can be formulated as in the following definition.

Definition (FAP - Fault Accommodation Problem)

Solve

$$P_{c_f}^{fa} \triangleq \langle O_n, \hat{S}_f, \hat{\theta}_f, \mathscr{U} \rangle \,, \tag{3}$$

where $(\hat{S}_f, \hat{\theta}_f)$ is the estimate of the actual constraints, e.g., provided by fault diagnosis algorithms. $\square$

Explicitly, it means to adapt only the controller parameters to the dynamical properties of the faulty plant, while the input and output of the plant used in the control-loop remain the same as for the fault-free case.

Preliminaries I

However, seen the way in which SRP or FAP propose a solution to the FTCP, note that the set of feasible pairs $(S_f \times \Theta_f)$ depends on the fault, where $\times$ defines the Cartesian product.

Fault recoverability I

Hence, the fault recoverability property may be seen as in the following definition.

Definition (Fault Recoverability)

A system is recoverable from a fault if a solution exists to at least one of the problems

$$\langle O_n, \hat{S}_f, \hat{\theta}_f, \mathcal{U} \rangle \text{ or } \langle O_n, \tilde{S}_f, \tilde{\theta}_f, \mathcal{U} \rangle \,. \tag{4}$$

□

If such a feasible pair does not exist, this means that O_n can be achieved neither by fault accommodation nor by system reconfiguration and the only possibility is thus to change the objective O_n to O_f.

Fault recoverability II

Anyhow, the choice of a new objective can be made autonomously in rare cases only. Most commonly, human intervention is needed using decision support from the diagnosis system regarding the overall goals for the plant. It should be noted that the choice of O_f could be made to include the fail-to-safe condition where control is no longer active but plant safety is not at stake.

That is to say, fault tolerant systems could be characterized also as fail-operational whether they continue their operation despite any singular fault or they may be considered as fail-safe if, after a serious fault, they are able to be conducted to an overall fail-safe position. In terms of this characterization, appropriate software simulations should be done before any physical implementation.

System supervision I

Regarding this, it is seen that there is not a simple task to design and to implement a fault tolerant control system; at any rate, it is still possible to design a good control action by systematic planning altogether with right and logic programming of the control actions in the overall software system.

Such a systematic design strategy assumes to make the analysis of fault propagation based on the Fault Management and Effects Analysis (FMEA) matrix, to consider the severity assessment of possible faults in terms of probabilities, to do the structural analysis for checking if the redundancy degrees of different subsystems are fairly distributed, and to provide the supervisory logic in order to plan the correct counteraction for a specific faulty situation.

In this respect, the most general problem is defined by the quadruple $\langle \mathcal{O}, \mathcal{S}, \Theta, \mathcal{U} \rangle$, where $\mathcal{O}$ is also a set of possible control objectives.

In view of its practical interpretation, $\langle \mathcal{O}, \mathcal{S}, \Theta, \mathcal{U} \rangle$ is defined as a system supervision problem in which the system goal is not predefined but has to be determined each time by taking into account the actual system possibilities.

System supervision II

Hence, the System Supervision Problem (SSP) may be described as in the following definition.

Definition (SSP - System Supervision Problem)

Monitor the triple $\langle O_n, S_f, \theta_f \rangle$ to determine whether the control objective is achieved; if this is not the case and the fault tolerant control problem does not have a solution, then find a relaxed objective $O_f \in \mathcal{O}$ and a pair $(\tilde{S}_f, \tilde{\theta}_f) \in (S_f \times \Theta_f)$ such that the system supervision problem

$$P^{ss}_{c_f} \triangleq \langle O_f, \tilde{S}_f, \tilde{\theta}_f, \mathcal{U} \rangle \tag{5}$$

has a solution. $\square$

Therefore, the system supervision problem may be described literally as the ability to monitor whether control objectives are met; if not, obtain / calculate a revised control objective and a new control structure and parameters that make a faulty closed-loop system meet the new modified objectives. Supervision should take effect if faults occur and it is not possible to meet the original control objective within the fault-tolerant scheme.

System reliability I

In any event, if no such triple $\langle O_f, S_f, \theta_f \rangle$ exists, failure is unavoidable.

This can be a design error or a deliberate choice to accept certain failure scenarios, e.g., for reasons of cost / benefit or small likelihood for a certain event.

Thus, the system reliability property may be seen as in the following definition.

Definition (System Reliability)

The system reliability defines the ability of the fault tolerant control system to complete a task satisfactorily, dealing with the period of time over which that ability is retained. □

In what follows, the characterization and the classification of the fault tolerant control methods are given.

Classification of fault tolerant control approaches I

Fault Tolerant Control (FTC) methods may be classified in two categories, namely passive (PFTC) and active (AFTC) approaches, as follows:

- Passive Fault Tolerant Control (PFTC): it assumes to have a fixed structure controller that must be considered since the controller design stage, whereas controllers are designed to be robust against uncertainties and some restrictive faults.

- Active Fault Tolerant Control (AFTC): it presumes to have a controller potentially able to deal with a large number of fault scenarios, which can also deal with a certain number of unforeseen or unexpected faults; they are provided with explicit fault detection / diagnosis schemes but the real challenge is real-time decision-making.

Implementation of fault tolerant control approaches I

Regarding the two classes of FTC methods presented before, the Fault Tolerant Control Systems (FTCS) may be implemented as passive (PFTCS) and active (AFTCS) systems, being defined as follows:

- Passive Fault Tolerant Control Systems (PFTCS): it is known that these systems are simple to implement, but, by using them, it is difficult to account for large number of fault scenarios, since they are unable to deal with unforeseen or unexpected faults; in fact, PFTC methods are robust control techniques that are suitable for certain types of structural faults, if these can be modeled as uncertainty regions around a nominal model; in this case, the PFT controller still provides the system with satisfactory stability and performance guarantees as long as such faults do not force the system step outside the stability radius; however, any controller with sufficiently large stability radius to cover most fault situations may become too conservative and still there is no guarantee that unanticipated or multiple faults can be handled; it would also require a large storage capacity for extensive faulty situations modeling and dedicated control solutions.

- Active Fault Tolerant Control Systems (AFTCS): it is evident that these systems are more complex to implement than the PFTCS and differentiate themselves from the first in that they take fault information explicitly into account; since there still exist different types of faults which cannot be adequately modeled as uncertainty, such as actuator or sensor faults, the AFT control design addresses this situation actively; AFTCS react to the system component faults either by system reconfiguration or fault accommodation control actions such that stability and acceptable performance of the entire system can be maintained, while degraded performance may have to be accepted in certain circumstances.

Characterization of active reconfiguration methods I

In order to design an automatic control system that provides safety in critical operation, this must be thought of as integrating at the design phase an intelligent mechanism capable of self-repairing in the impaired situation.

At any rate, in the case of different impaired situations, the active fault tolerant control methods are more or less based on Fault Detection and Diagnosis (FDD) methods, indicating the subsystem affected by the fault and thus giving an idea about the actual configuration; therefore, the AFTC methods are based on the information they process from the fault-tolerant instrumentation and software fault-diagnosis automatic procedures. In this respect, it is desired that the automatic implementation of several methods, capable to measure the effectiveness of the current control mode, to identify the controls which have been lost when the measures exceed their safety thresholds, to determine the characteristics of the remaining controls, and to redesign the control laws for the remaining active controls, respectively, all without the intervention of a control engineer.

Thus, the implementation of different AFTC methods capable of self-repairing or self-reconfiguration may have different views depending on several factors, e.g., either the mathematical tools or the design approaches used, or the way achieving reconfiguration. In such control systems, the controller compensates for the effects of faults either by selecting a precomputed control law (projection-based methods) or by synthesizing a new control scheme (online automatic control redesign methods).

Questions? vs. Answers!

Acknowledgments:
Marcel Staroswiecki, Dumitru Popescu,
Andrei Sperilă, Sabin Diaconescu, Teodor Rotaru,
Nicolai Christov, Cristian Oară,
Theodor Nicu.